CERTAIN-COAST — Technical annex

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast

How the methods work, what was tested before the project, and what is new. This annex supplements the fields "How will the project reach the planned end state?" and "Work-plan" of the application form; work packages (WP), tasks (T), deliverables (D), months and milestones (MS) are those of the form. Test results come from synthetic test systems and preparatory models, not from measurements made by the project at the sites.

1. Two sites, three coupled hazards

The project protects two sites on the Albanian Adriatic coastal plain: at Durrës, the wellfields that supply drinking water and the port; near Fier, the landfall of the Trans Adriatic Pipeline (TAP). Both stand on low ground near active faults, and at Durrës the land is sinking; the earthquake of 26 November 2019 caused liquefaction near the coast (Mavroulis, Lekkas and Carydis, 2021). Three hazards threaten these assets, and they are coupled through two variables, the relative sea level and the water table. Sea level enters the aquifer at its seaward boundary and sets how far seawater advances; the water table sets the effective stress in the sands and so their resistance to liquefaction; subsidence raises the relative sea level at every asset. Pumping lowers the water table: it increases seawater intrusion and, below the preconsolidation head, permanent subsidence, but it raises the factor of safety against liquefaction. No hazard can be managed alone, so one indicator is defined per hazard, oriented so that larger means more dangerous (Table 1).

Table 1. Hazard indicators, thresholds and lead times.

Hazard (assets)	Indicator	Threshold	Lead time, cadence
Seawater intrusion (Durrës wellfields)	I_1: chloride at each production well	Agreed with the utility and AMBU (for example 250 mg/L)	Weeks to months; weekly
Liquefaction (Durrës port; TAP landfall)	I_2: liquefaction potential index under the design shaking at the current water table	Class boundary agreed with the operators	Daily; recomputed at once after an earthquake
Extreme water level (all assets)	I_3: total water level at each asset	Design or operational level of each asset	Hours to days for storms; return levels yearly

An alarm is missed when the indicator exceeds its threshold within the lead time and no alarm was issued; the long-run miss rate is the quantity the project certifies. Earthquakes are not forecast: the liquefaction alarm states how the ground would respond if the design earthquake struck now, a state that changes with the water table.

2. The method chain

Six steps turn data into bulletins (Figure 1). Each step has a pass test and a fallback (Section 7). Three loops close the chain: verified outcomes update the alarm margins, new data are assimilated into the model, and the certified safe window of Section 6 tells the network how many years of safe operation each sensor adds.

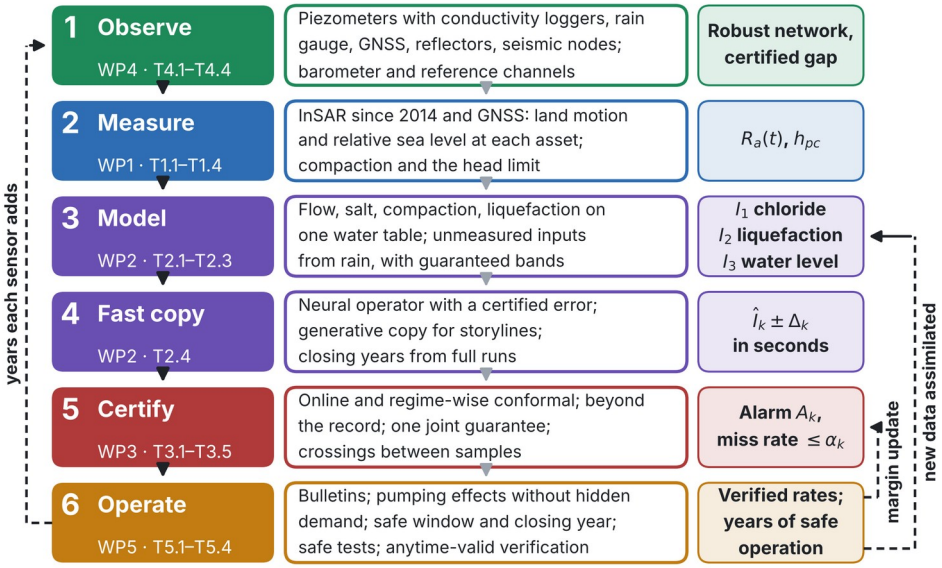


Figure 1. The CERTAIN-COAST method chain for the Durrës wellfields and port and the TAP landfall near Fier. Solid arrows: from data to bulletins. Dashed arrows: the three feedback loops.

3. Observe and measure (WP1, WP4)

3.1 Relative sea level at each asset (WP1, T1.1–T1.4)

Sentinel-1 ascending and descending time series since 2014 give the vertical and east components of land motion; a trajectory model separates trend, season, the coseismic step of 26 November 2019 and the postseismic transient. Corner reflectors and GNSS at the assets (T1.2) give stable radar targets on soft ground and tie the solution to a global frame. The tide gauge records sea level η_{TG} relative to its own benchmark, which also moves, so the relative sea level at asset a adds the differential vertical land motion u_z, propagated with its full covariance (target vertical-velocity uncertainty 1–2 mm/yr; Ohenhen et al., 2023):

$$R_a(t) = \eta_{\mathrm{TG}}(t) + u_z^{\mathrm{TG}}(t) - u_z^{a}(t) \tag{1}$$

Elastic and inelastic compaction fitted to the InSAR and head records give the head limit h_{pc} above which pumping causes no new permanent subsidence (Lees, Knight and Smith, 2022). The same $R_a(t)$ drives all three hazard models, so the warnings cannot disagree about where the sea is.

3.2 A network designed for the warnings, which corrects its own sensors (WP4, T4.1–T4.4)

The network is designed before it is bought (D4.1, Month 6). Candidates are piezometers with conductivity loggers, a local rain gauge, GNSS, corner reflectors, seismic nodes and, where access allows, fibre-optic sensing; the monitoring wells of AMBU (existing wells only) are candidate sites. Each candidate is valued by goal-oriented Bayesian design (Attia, Alexanderian and Saibaba, 2018) in two currencies: the guaranteed width it removes from the warning bands, and the years of certified safe operation it adds (Eq. (10)). A sensor is bought only if its guaranteed value is positive: in a preparatory test, a tide gauge that a standard ensemble rated as narrowing the recharge band by 15% widened the guaranteed band by 16–21%.

Loggers respond to air pressure (Rasmussen and Crawford, 1997) and drift. A barometer and a sealed reference channel on each logger hub are therefore design candidates beside the piezometers; their errors are removed by marginalisation, and the network is chosen against the worst plausible error model, with a certified distance from the best network (robust design; Chowdhary, Attia and Alexanderian, 2025). In a preparatory test with 12 sensors the worst-case information was 5.59 nats against 3.67 for the classical design (+52%), with a certified optimality gap of 18%; 90% forecast intervals covered 88–93% of outcomes after correction and 49–62% without, and corrected heads were within 2.9 cm root-mean-square against 7.6 cm raw.

At the TAP landfall, T4.2 installs one GNSS station, two corner reflectors, a seismic station and a water-table logger under TAP's security rules. Ambient seismic noise tracks the water table (Clements and Denolle, 2018), also on unused telecommunication fibre (Rodríguez Tribaldos and Ajo-Franklin, 2021); once calibrated against piezometers (T4.4), velocity changes measure the water table between wells at the port and the landfall, and a drop after an earthquake measures the softening of the ground.

4. Model and certify (WP2)

4.1 One coupled model (T2.1–T2.3)

One model, set up for the two sites on one coastal motion field (T2.1), couples variable-density flow, salt transport with memory in clays, the sea as a moving boundary driven by $R_a(t)$, compaction, the total water level at each asset, and liquefaction triggering with the water table taken from the flow model at every time step (Seed and Idriss, 1971; Boulanger and Idriss, 2014). Groundwater rise under sea-level rise is a hazard in its own right (Befus et al., 2020) and raises liquefaction severity where water tables are shallow (Grant et al., 2021); here one water table drives all three hazards, so a pumping decision is checked against all of them at once. Parameters are estimated jointly from heads, conductivity, chloride, InSAR and seismic velocity by an ensemble smoother (Emerick and Reynolds, 2013); the model is differentiable, so adjoint gradients of every indicator are available (Shen et al., 2023). Priors are stated so that an expert can read and dispute them (Simpson et al., 2017; Fuglstad et al., 2019) and are checked after calibration for prior-data conflict (Evans and Moshonov, 2006).

4.2 Guarantees for the inputs and boundaries nobody measures (T2.3)

Recharge, vadose delay, storage, the sea-level and inland boundaries, the initial head and the dispersion width are never measured, yet every warning depends on them. They are reconstructed from rain and heads by a function-space diffusion prior with an ensemble update through the exact solver, with the errors of the rain product simulated in the prior. Their bands are calibrated by conformal risk control (Angelopoulos, Bates, Fisch, Lei and Schuster, 2024) on simulated aquifers that carry these errors, and checked on measured anchor periods, so the guarantee does not depend on the simulator. With a loss $L_i(\lambda)$ between 0 and 1 for calibration case i and its mean $\hat{R}_n(\lambda)$ over n cases:

$$\hat{\lambda} = \inf\left\{\lambda : \frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1} \le \alpha\right\} \text{ and } \mathrm{E}\left[L_{n+1}(\hat{\lambda})\right] \le \alpha \tag{2}$$

For α = 10%, at least 9 anchor periods give the average guarantee and 22 a 90% certificate. Under drift the calibration is matched to the forecast horizon, and a computable floor shows how far each band is from the narrowest valid one. In the preparatory test the raw ensemble missed 17–20% of the weeks and the calibrated bands 9.5–10% (target 10%); the risk stayed at 0.003–0.082 up to 20 years ahead under drift.

4.3 A fast copy with a certified error (T2.4)

Warnings need thousands of model runs a day. A neural operator (Li et al., 2021; Lu et al., 2021) is trained on runs of the coupled model, and its error on each forecast is bounded conformally (Ma et al., 2024), with the score normalised by the goal-oriented dual-weighted residual η_k (Becker and Rannacher, 2001), so the bound is tight where the copy is right about the quantity that triggers the alarm:

$$\Delta_k(\xi) = \hat{q}_{1-\delta}\,\eta_k(\xi) \text{ and } Pr\left[\left|I_k(\xi) - \hat{I}_k(\xi)\right| \le \Delta_k(\xi)\right] \ge 1-\delta \tag{3}$$

Forecasts whose bound is too wide are recomputed with the full model on a reduced domain. A generative copy (Gao, Kaltenbach and Koumoutsakos, 2024) explores long storylines of sea-level rise, subsidence, droughts and pumping and chooses which of them the full model runs; in a preparatory test its statistics alone put the closing year of the safe window 2–8 years too late, so closing years are always certified from full runs with the exact binomial bound (Clopper and Pearson, 1934).

5. Certified alarms (WP3)

5.1 Alarm rule, calibration under shift and beyond the record (T3.1–T3.2)

Thresholds τ_k and tolerable miss rates α_k are agreed with the operators (T3.1). Alarm k fires when the forecast at lead time L_k comes within a margin m_k of its threshold. The margin adds the certified copy error to a forecast error γ_k that rises after a miss and falls slightly otherwise; with a constant step λ the long-run miss rate stays within $C_k/(\lambda T)$ of its target for any sequence of data (Gibbs and Candès, 2021, 2024; Angelopoulos, Barber and Bates, 2024):

$$A_k(t) = 1\left(\hat{I}_k(t+L_k) \ge \tau_k - m_k(t)\right) \text{ and } m_k(t) = \Delta_k(t) + \gamma_k(t) \tag{4}$$

$$\gamma_k(t+1) = \gamma_k(t) + \lambda_t\left[\mathrm{miss}_k(t) - \alpha_k\right] \text{ and } \left|\frac{1}{T}\sum_{t=1}^{T}\mathrm{miss}_k(t) - \alpha_k\right| \le \frac{C_k}{\lambda T} \tag{5}$$

The guarantee also holds within each regime (drought, storm season, after an earthquake, each asset) by conformal prediction with conditional guarantees (Gibbs, Cherian and Candès, 2025), and for events rarer than the record by extreme conformal prediction (Pasche, Lam and Engelke, 2026). Still-water extremes use a generalised Pareto tail (Coles, 2001) whose threshold moves with the measured relative sea level of Eq. (1). In preparatory tests on a coupled variable-density model, an extreme-aware margin kept the share of seasons above the chloride limit at 4–9% across drought, human-feedback and compound storylines, where a static alarm allowed 13–21%.

5.2 One guarantee for the three assets (T3.3)

The alarms share inputs, so their misses are dependent. The probability that any alarm is missed is bounded by the sum of the tolerances, which are allocated where tightening them costs least in consequence-weighted false alarms, within caps agreed with the operators. The margins are calibrated together by Learn then Test

(Angelopoulos, Bates, Candès, Jordan and Lei, 2025), so every certified rate holds at once with probability at least $1-\beta$. Civil protection receives one joint guarantee for the wellfields, the port and the landfall.

5.3 Warnings for what happens between samples (T3.2)

Chloride is sampled weeks or months apart. Sparse samples pin the distribution of chloride on the sampling dates but not its time scale, so a well can turn salty between two samples that both look safe. The warning is therefore the probability that chloride crosses its limit $c^{\star}$ at any time in the season T, reported as a band over the set M of all models consistent with the records, in the spirit of robust bounds over calibrated models (Gierjatowicz et al., 2020) and of online forecasts from irregular records (Krach, Nübel and Teichmann, 2026):

$$\underline{p}=\inf_{P\in M} P\left[\max_{t\in T} c(t)>c^{\star}\right] \text{ and } \overline{p}=\sup_{P\in M} P\left[\max_{t\in T} c(t)>c^{\star}\right] \tag{6}$$

The band is widened by a conformal margin calibrated on simulated aquifers and anchor periods and issued online as the conditional probability given the irregular samples and the daily heads; the network design adds the observation that closes it (a logger, or samples paired 1, 3 and 7 days apart). In the preparatory test the chance of a crossing within the season was 0.78, against 0.023 on a sampling date (34 times); the band was 0.18–1 with quarterly samples and 0.76–0.80 with a daily logger; for skewed chloride the margin restored 90.3% coverage (74% without), and a head logger raised the warning skill from 7% to 20%.

5.4 Back-testing and verification in operation (T3.4–T3.5)

The chain is replayed over the record since 2014 with the system frozen at each issue time, under a protocol published with a DOI before the results (D3.2, Month 27). At 95% confidence, verifying a 10% miss rate needs 29 exceedance events with no miss, 46 with one or 61 with two; a 5% rate needs 59, 93 or 124. In operation the ledger is published every quarter, so each miss rate is looked at many times, and a fixed-sample bound recomputed at every look loses its guarantee: in a simulation of six calibrated alarms it flagged at least one as broken in 68% of runs. Verification therefore uses e-processes, which stay valid however often they are checked (Ville, 1939; Vovk and Wang, 2021; Ramdas et al., 2023), with $x_i=1$ if event i was missed:

$$E_k(n)=\sum_j \pi_j \prod_{i=1}^{n} \left(\frac{q_j}{\alpha_k}\right)^{x_i}\left(\frac{1-q_j}{1-\alpha_k}\right)^{1-x_i} \text{ and } Pr\left(E_k(n)\geq 1/\beta \text{ for some } n\right)\leq \beta \tag{7}$$

In the simulation the e-process flagged a calibrated alarm in 2.2% of runs (at most 5% is guaranteed) and detected an alarm whose miss rate had risen to 25% in 84% of runs within 100 events.

6. Operate and decide (WP5)

Twelve months of live bulletins (T5.2) follow a dry run (T5.1). Each bulletin states, per asset, the forecast, the margin, the alarm state, the certified miss rate and the distance to the closing of the safe window, and is logged with its inputs and code version so that it can be reproduced and audited. A post-event protocol agreed with civil protection (T5.4) remeasures heads, chloride, GNSS and seismic velocity within days of a strong earthquake or storm, updates the model and recomputes the alarms and the window.

6.1 The effect of pumping without the hidden drought demand (T5.3)

In droughts the utility pumps more while recharge falls, so a hidden cause drives both the pumping and the salinity, and substitutes for hidden confounders inferred from the treatment history (Bica, Alaa and van der Schaar, 2020) can miss it. CERTAIN-COAST measures the confounding path instead: the heads, after the drawdown measured in the pumping tests is added back, read the recharge, and a calibrated physics check rejects any substitute that misses a confounder before an effect is reported. In the preparatory test the bias of the summed well effects was 0.59 unadjusted, 0.56 with a history substitute and 0.00 with the heads; the check had a size of 6% and a power of 99–100%. On the calibrated chloride model of Fushë-Kuqe, 40% less recharge would grow the shallow area above 250 mg/L from 31 to 65 km² (effect +33 km², conformal band 15–49 km²).

6.2 The certified safe operating window and the year it closes (T5.3)

Pumping moves the hazards in opposite directions, and sea-level rise and subsidence tighten all of them. At each bulletin and over the adaptation horizon the project computes the set of decisions u (wellfield pumping Q_w, port dewatering) that meets the demand D, keeps every hazard within its certified limit and causes no new permanent subsidence, and the relative sea level and year at which this set becomes empty:

$$W_\alpha(R)=\left\{u: Q_w \geq D,\ \Pr_\theta\left[I_k(u,R;\theta)>\tau_k\right]\leq \alpha_k\,(k=1,2,3),\ h(x,u,R)\geq h_{\mathrm{pc}}(x)\right\} \tag{8}$$

$$R^\star_\alpha=\text{largest } R \text{ with } W_\alpha(R)\neq\varnothing \text{ and } t^\star_\alpha=\text{first } t \text{ with } W_\alpha(R(t))=\varnothing \tag{9}$$

This is a certified, multi-hazard adaptation tipping point (Kwadijk et al., 2010; Haasnoot et al., 2013); tipping points exist for seawater intrusion alone (Mazi, Koussis and Destouni, 2013). At the closing point the least dewatering that keeps liquefaction within its limit, A, meets the most dewatering the wellfield tolerates, B, and one multiplier μ converts any change of a margin, from a sensor, a better model or a pumping test, into years:

$$\mu=-\left[\frac{d(B-A)}{dt}\right]^{-1}_{t=t^\star_\alpha} \text{ and } \Delta t^\star_\alpha \approx \mu(\Delta B-\Delta A) \tag{10}$$

On a synthetic transect with the three hazards of Durrës and illustrative values, the certified window closed at a relative sea level of 0.41 m, after 38 years; plans that watch one hazard believed they had 43 to 97 years, and the margins cost 84 years against the window without them. Valued in years by preposterior analysis, with Eq. (10) as a first-order shortcut, a sentinel piezometer between the wellfield and the sea added 21.6 years, and a GNSS station and two corner reflectors 5.6 and 5.0 years, where the classical criterion values them at zero because they narrow none of today's forecasts.

6.3 Safe pumping tests chosen for the years they add (T5.3)

Routine pumping keeps the water table in a narrow band, so the records say little about how chloride responds near the onset of intrusion, which decides when the window closes. Short pumping and dewatering tests, each approved by the utility, approach that onset safely. The response is linear in unknown coefficients of hinge features of the head, with a confidence set that holds in every week (Abbasi-Yadkori, Pál and Szepesvári, 2011). A test is chosen for the years it adds to the certified closing year, as in active identification with guarantees (Mania, Jordan and Recht, 2022) but for the decision, and only if the upper confidence bound of chloride stays below a test limit in every week of the test and the recovery (Koller et al., 2018). In the preparatory test the campaign raised the certified closing year from 10.8 to 31.9 years (true value 45) with chloride below 123 mg/L in 95% of runs, whereas tests chosen for parameter accuracy alone passed 250 mg/L in 42% of runs; the certificate held in all 1,000 runs. The years rest on the prior from the calibrated model, so the certified year is reported for the prior error radius S and for $2S$ (doubling it cost 17.3 of the 31.9 years), for readers who trust the model less (Andrews and Shapiro, 2021).

6.4 Pumping rule and the value of the warning for each user (T5.3, D5.1)

Within the window, the pumping and drought rule supplies as much water as possible; it is checked for a sudden pumping stop, after which the water table recovers and the factor of safety falls, computed with the fast copy, confirmed with the full model and agreed with the utility and AMBU. Because each miss rate is certified, the value of a warning to each user in the cost-loss model (Murphy, 1977; Richardson, 2000) has a guaranteed lower bound for every cost-loss ratio agreed in T3.1, reported in D5.1.

7. Gates, pass tests and fallbacks

Table 2. What each step must achieve before the next starts, and what happens if it does not.

Step and gate	Pass test	Fallback
Network design (WP4; MS1, Month 6)	Approved by the steering committee and the operators; certified optimality gap of at most 25%	Design re-run with the remaining budget; seismic nodes if fibre is not available
Network and coastal motion (WP4, WP1; MS2, Month 12)	At least 90% data availability; corrected heads within 3 cm root-mean-square of manual dips; vertical-velocity uncertainty of about 1–2 mm/yr at asset points	Corner reflectors and scatterers on structures where radar coherence is lost

Step and gate	Pass test	Fallback
Coupled model and unmeasured inputs (WP2; MS3, Month 18)	Held-out heads, chloride, InSAR and seismic velocity inside the 90% predictive intervals at close to the nominal rate; band for crossings at most 0.25 wide for each wellfield	Model built in stages; two-dimensional sections through each asset with a wider error bound; a wider simulation-calibrated band, reported as such, if anchors are too few
Certified copy and alarm prototype (WP2, WP3; MS4, Month 22, go/no-go)	Error bound met on test runs; the prototype passes a dry run with the operators	Full model on a reduced domain for flagged forecasts
Back-testing (WP3; MS5, Month 27)	Each miss rate at or below its target with a 95% confidence bound, crossings between samples included, under the pre-registered protocol	Margin widened: the miss target is kept and the extra false alarms are reported
Live operation (WP5; MS6, Month 34)	Twelve months of logged bulletins; no e-process reaches 1/β; safe window reported for each asset; every pumping test below its chloride test limit	If no strong event occurs, verification rests on back-testing and storylines; smaller and shorter tests, or the window from routine records
Hand-over (WP6, WP7; MS7, Month 36)	Open software and data with a DOI reproduced outside the team (D6.1, Month 33); operating procedures adopted (D6.2)	—

8. What is new beyond the state of the art

The alarms assemble published methods of 2021–2026 (online and conditional conformal prediction, extreme conformal prediction, Learn then Test and anytime-valid verification); their joint use for coupled coastal hazards is an application and is not claimed as new. Six contributions were derived and tested before the project (Table 3); the novelty statements rest on our literature search to date.

Table 3. The six new contributions, the result of their preparatory test and where they are used.

Contribution	What is new	Preparatory test	Where
(i) Guarantees for quantities nobody measures	Bands for recharge, vadose delay, boundaries and dispersion that stay valid when the simulator or the rain product is wrong, with a computable floor	Raw ensemble missed 17–20% of weeks; calibrated bands 9.5–10% (target 10%)	T2.3; D2.1, MS3
(ii) The value of a sensor	The guaranteed band a sensor removes and the years of safe operation it adds	GNSS and corner reflectors: 5.6 and 5.0 years, where the classical criterion gives zero; a tide gauge rated useful widened the band by 16–21%	T4.1; D4.1, MS1
(iii) Pumping effects free of hidden confounding	The confounding path measured by the heads, with a calibrated check of every substitute	Bias 0.00 with the heads, against 0.56–0.59; check size 6%, power 99–100%	T5.3
(iv) Certified safe window and safe pumping tests	One window for three hazards and the year it closes; tests chosen for the years they add, with a chloride safety filter	Window closes after 38 years where single-hazard plans assume 43–97; tests: 10.8 to 31.9 years, chloride below 123 mg/L in 95% of runs	T5.3; D5.1, MS6
(v) A network that corrects its own sensors	Barometer and reference channels placed by design; robust choice with a certified optimality gap	+52% worst-case information; gap 18%; interval coverage 88–93% against 49–62%	T4.1–T4.3; D4.1
(vi) Warnings for what happens between samples	A certified band for a crossing at any time in the season, and the observation that closes it	Crossing 0.78 against 0.023 on a sampling date (34 times); band 0.18–1, narrowed to 0.76–0.80 by a logger	T3.2; D2.1, D3.2

Together they answer questions that no current coastal warning answers for its operators: how often a warning will be missed and so what silence means; how many years of safe pumping and dewatering remain as the sea rises and the land sinks; and which sensor or pumping test buys more of them. The system starts at TRL 3, each component shown in preparatory tests, and ends at TRL 6, demonstrated on the operators' own sites through twelve months of live operation with the water utility, AMBU, the port authority, TAP and civil protection.

Software and data. The model, fast copies, alarms, safe window, test planner and network design are released under an open licence with containers and tests (D6.1). Operator data are used under the access agreements of T7.2 and the security rules of each operator; derived products, the bulletin log and the verification ledger are released openly following FAIR principles, and the preparatory tests are scripted and released with the software.

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